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Design and preliminary test results of the charge sensitive amplifier for gain-less charge readout in high-pressure TPC

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Abstract: This paper presents the design and electrical test results of a low-noise front-end chip (named Topmetal-S) in a High-pressure Time Projection Chamber (TPC) for searching the neutrinoless double beta decay. The Topmetal-S has been fabricated in a 130 nm CMOS technology. The proposed front-end chip consists of a charge collection electrode, a Charge Sensitive Amplifier (CSA) and peripheral circuits. The test results indicate that the CSA features an input linear dynamic range of approximately 6.64 fC, a charge-conversion gain of about 220 mV/fC and an Equivalent Noise Charge (ENC) of approximately 115 e^- after a digital trapezoidal pulse shaper.

Keywords: Analogue electronic circuits; CMOS readout of gaseous detectors; Front-end electronics for detector readout

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1 Introduction

Neutrinoless double beta ($0\nu\beta\beta$) decay stands as one of the most critical frontiers in particle physics. Its observation would not only confirm neutrinos as Majorana particles but also exert a profound influence on the field of particle physics. To detect such a rare process, various strategies have been explored and developed in $0\nu\beta\beta$ decay experiments, such as those involving ^{76}Ge in GERDA [1], MAJORANA [2], and CDEX [3], ^{100}Mo in CUORE [4] and CUPID [5], and ^{136}Xe in KamLAND-Zen [6], EXO [7], NEXT [8], and PandaX [9]. Given the extremely low rate of $0\nu\beta\beta$ decay, achieving zero background is paramount to enhance the sensitivity of these experiments.

In response to this challenge, an innovative concept was proposed in [10], searching for $0\nu\beta\beta$ decay of ^{82}Se using a high-pressure gas time projection chamber (TPC) with $^{82}SeF_6$ as the working medium. The Q value of ^{82}Se is exceptionally high at 2.996 MeV, surpassing that of most natural radioactive backgrounds. The signal and background are distinguished using event topology in TPC, which can greatly reduce the experimental background. To materialize this concept, a new experiment called NνDEx-100 (No neutrino Double-beta-decay Experiment) utilizing 100 kg of natural SeF_6 gas is being developed. As the core detector, the TPC is positioned within the barrel part of the pressure chamber, as shown in figure 1(a). It comprises an insulating layer, a high-voltage plane, a field cage, and a readout plane. When $0\nu\beta\beta$ occurs, ^{82}Se in SeF_6 gas releases two electrons. Due to the electronegativity of SeF_6, those electrons swiftly form negative ions with surrounding SeF_6 molecules. As the $SeF^-_{5,6}$ ions reach the readout plane, their signals are read out. Since the drift velocities of SeF^-_5 and SeF^-_6 ions are different, their arrival times vary, allowing for the calculation of the drift distance. To achieve a sharp peak at $Q_{\beta\beta}$ with a precision of 1%, a charge sensor called Topmetal is designed to collect ions directly, shown in figure 1(b). The readout plane shown in figure 1(c) comprises the focusing layer and the readout electronics layer composed of directly charge collection sensors named Topmetal-S. The focusing layer generates a specific electric field structure that allows the drifting charges to pass through small holes, subsequently being collected with 100% efficiency at the $\sim 1\,mm^2$-sized readout

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electrodes on the surface of the Topmetal-S chips. The TPC detector of NνDEx requires a meter-sized charge readout plane with around 10^5 Topmetal-S sensors on the plane without gas-electron avalanche. This innovative scheme eliminates conventional avalanche fluctuations but necessitates exceedingly low internal noise on the front-end amplifier to achieve sufficient energy resolution.

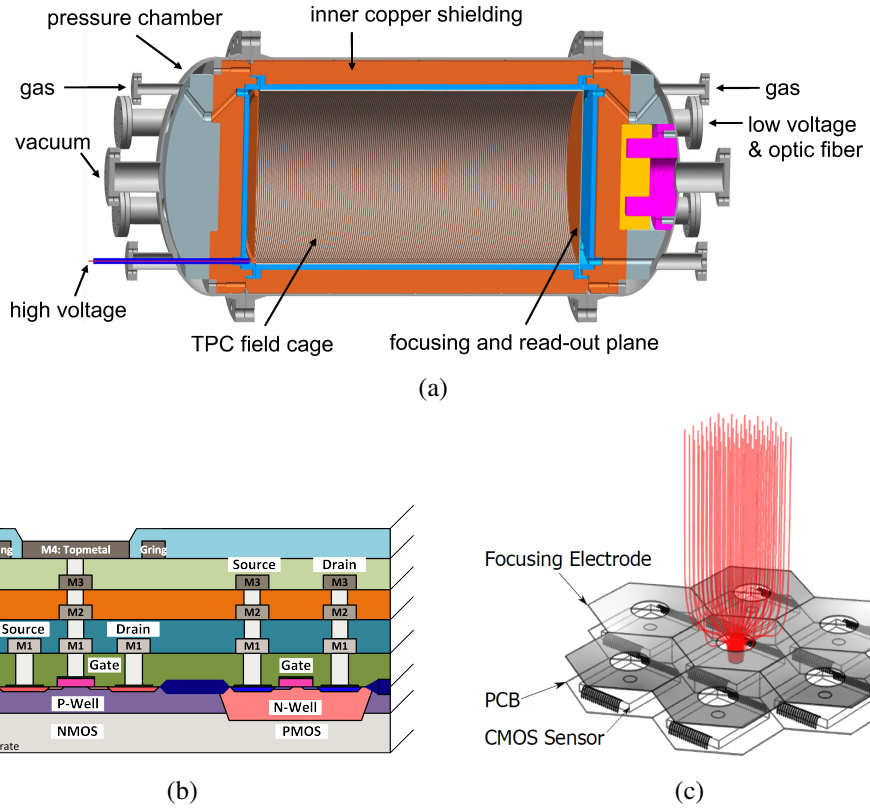


Figure 1. (a) The main body of NνDEx-100 [11]. (b) The conception of Topmetal. (c) The structure of the focusing and readout plane [11].

In this paper, the Topmetal-S sensor is presented to realize the low-noise collection of signals. The structure and detailed design of Topmetal-S are described in section 2. The electrical test results are presented in section 3, and the conclusion is drawn in section 4.

2 Design of Topmetal-S

2.1 Overall design

Topmetal-S in this version is composed of an exposure top-most hexagon metal with a diameter of 1 mm as a Charge Collection Electrode (CCE), a Charge Sensitive Amplifier (CSA) and peripheral circuits. The output analog signal of the CSA is transmitted off-chip by a unity-gain analog buffer. The total size of the Topmetal-S chip in this version is 2632 µm × 1794 µm.

2.2 Charge Sensitive Amplifier

The structure of the CSA is shown in figure 2. It is composed of a single-end folded cascode amplifier, a well-controlled feedback capacitor (C_f) and an n-type MOS transistor (M_f) with an adjustable-voltage

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gate. The CCE is directly connected to the CSA. A metal layer, named Guard-ring, encircles the CCE. In the experiment mode, the Guard-ring is connected to the ground to focus the electric field. In the electrical test mode, a test pulse can be injected into the Guard-ring to generate charge by the parasitic capacitor (C_{inj}) between the CCE and Guard-ring. C_d shown in figure 2 is the input capacitor and its value includes the parasitic capacitance of the CCE, the routing between the CCE and the CSA and the input transistor M1. C_f is implemented through two pieces of the metal layer. The decay time of the CSA could be adjusted up to several milliseconds by changing the value of VDIS. VREF is set to change the DC operating point of the CSA. A p-type transistor is chosen as the input device because it has a lower flicker noise contribution compared to an n-type transistor [12]. To optimize the noise, the size of the input transistor is designed to match the parasitic capacitance of the CCE.

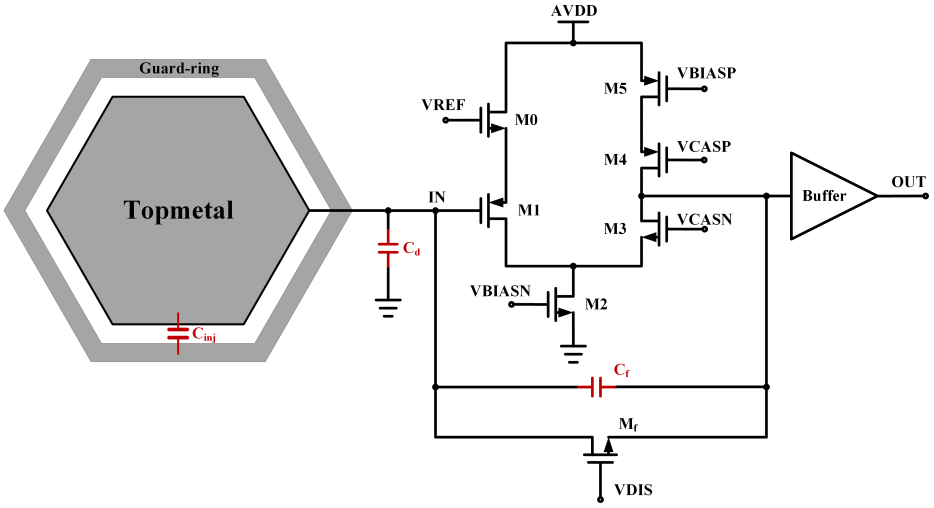


Figure 2. The structure of the CSA in Topmetal-S.

2.3 Peripheral circuits

All the bias voltages of the CSA can be provided by in-chip or off-chip digital-to-analog converters (DACs) by changing switches. The in-chip DACs are configured by a serial peripheral interface integrated into the test chip. The reference voltage of the DAC can also be supplied by an in-chip or off-chip bandgap. The routing of each bias is surrounded by the shielded metal layers just as a coaxial cable to reduce interference.

3 Test results

Topmetal-S has been designed and fabricated in a 130 nm CMOS process. As shown in figure 3(a), the core is composed of analog and digital parts with 51 pads. The power supply of the analog and digital circuits is 3.3 V and 1.2 V, respectively. The chip has been wire bonded on the test board shown in figure 3(b). Due to the demand for extremely low noise in NνDEx, analog and digital grounds are separated in the chip and the printed circuit board (PCB) designs. All the power and biases on the PCB are filtered by capacitors which vary from 0.1 µF to 10 µF. The electrical tests are carried out in a shielding chamber. Test pulses are injected on the Guard-ring. Figure 3(c) shows the output waveform

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through the analog buffer and the decay time is about 220 ms. The drift field of NνDEx-100 will be as high as 400 V/cm, corresponding to a drift velocity of negative ions above 20 cm/s [11]. Once $0\nu\beta\beta$ occurs, the arrival time of SeF_6^- on one Topmetal-S will be later than SeF_5^- of about 500 ms. Moreover, the neural network can further suppress the background by a factor of about 10, while keeping a 90% signal efficiency when event topology information is used. So, the final background level is 0.05 evts/yr in ROI (region of interest). After five years of operation, the background is about 0.25 events in ROI. Thus, NνDEx-100 is almost a zero-background experiment [11].

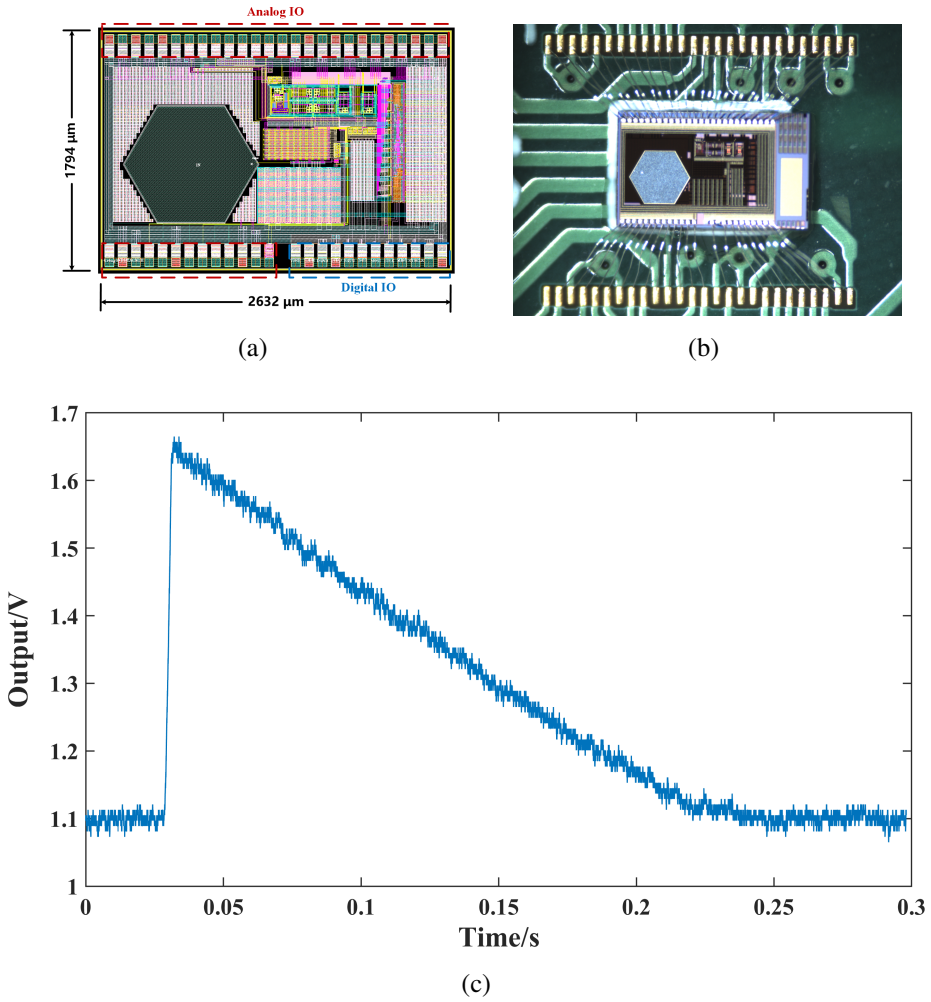


Figure 3. (a) Layout. (b) A Topmetal-S die bonding in PCB. (c) The response of the test chip.

3.1 Capacitors in Topmetal-S

The circuit comprises three crucial capacitors (C_d, C_f and C_{inj} shown in figure 2) that significantly impact the performance, including gain and noise characteristics. To determine the value of an unknown capacitor, CCE is bonded to the PCB. The output from the analog buffer is connected to an oscilloscope. One terminal of an accurate resistor is connected to the signal generator, the other is connected to the CCE. Hence an RC circuit is created, where the capacitance to the ground includes contributions from the CCE, the routing metal and the input transistor. By testing the cutoff frequency

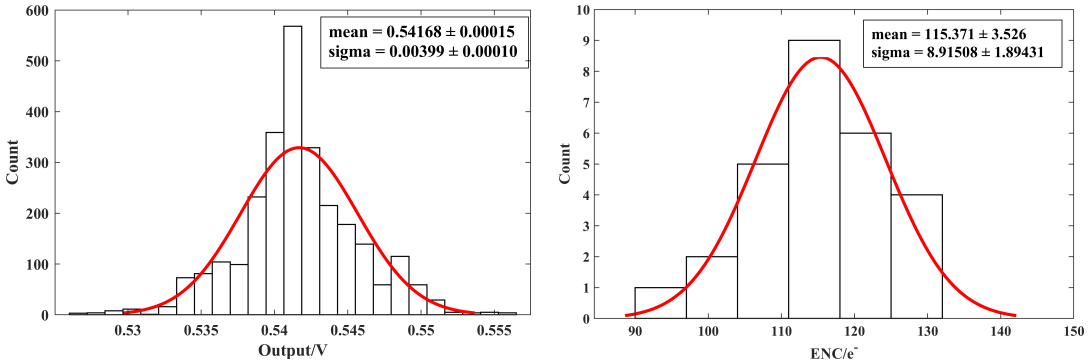
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of the RC circuit, the value of C_d is approximately 3 pF. A capacitor C_{test} with high precision is connected to the CCE to creat a bridge circuit on PCB. Two pulses with reversed phase are injected to the Guard-ring and C_{test}, respectively. By adjusting the amplitude of the injection, the amplitude of the output at CSA is zero, that is, the charge in C_{inj} equals to the one in C_{test}. C_{inj} is about 250 fF according to the test above. C_f is calibrated by a test chip integrated with two different feedback capacitors. The output signal amplitude is approximately equal to Q/C_f. With the same injection, the ratio of the output amplitude is the same as that of the capacitors. The relative size of the capacitance is determined by this method. The following test results are based on the above capacitance measurements.

3.2 Noise and stability

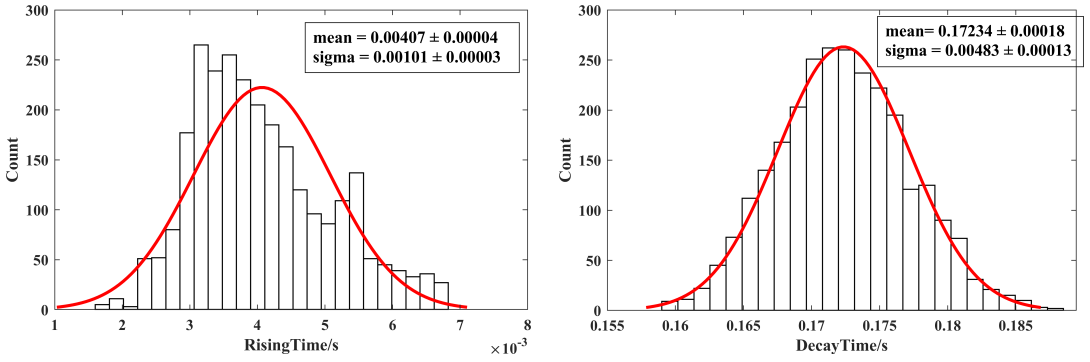
The Topmetal-S sensors will be used for the long-running process in NνDEx. It's necessary to verify the stability of the CSA. To assess the stability, a 10 mV square wave is injected into C_{inj} and then coupled to the CSA input node, resulting in an input charge of 15.6 ke$^-$. We analyze the output amplitude, noise, rising time, and decay time from 100 waveforms per hour. The noise performance of the Topmetal-S chip is characterized by the Equivalent Noise Charge (ENC).

The test results contain 2700 points collected in 27 hours, the distributions of some parameters of the output waveforms are shown in figure 4. The standard deviation (sigma) of the output amplitude, rising time (80% of the prominence height) and decay time (80% of the prominence height) is 0.00399 V shown in figure 4(a), 0.00101 s shown in figure 4(c) and 0.00483 s shown in figure 4(d),



(a) The distribution of output amplitude after filter.

(b) The distribution of ENC after filter.



(c) The distribution of rising time.

(d) The distribution of decay time.

Figure 4. Test results of stability and ENC.

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respectively. The ENC is about 115 e^- shown in figure 4(b). Based on the expected 1% FWHM energy resolution of NνDEx-100, ENC should less than 45 e^-. The sigma of ENC exhibits the variation of the noise per hour, which has not yet met the requirement. The noise of both the charge sensitive amplifier and the test system will be optimized to meet the requirement in the next version. The distribution of the baseline is also monitored in figure 5.

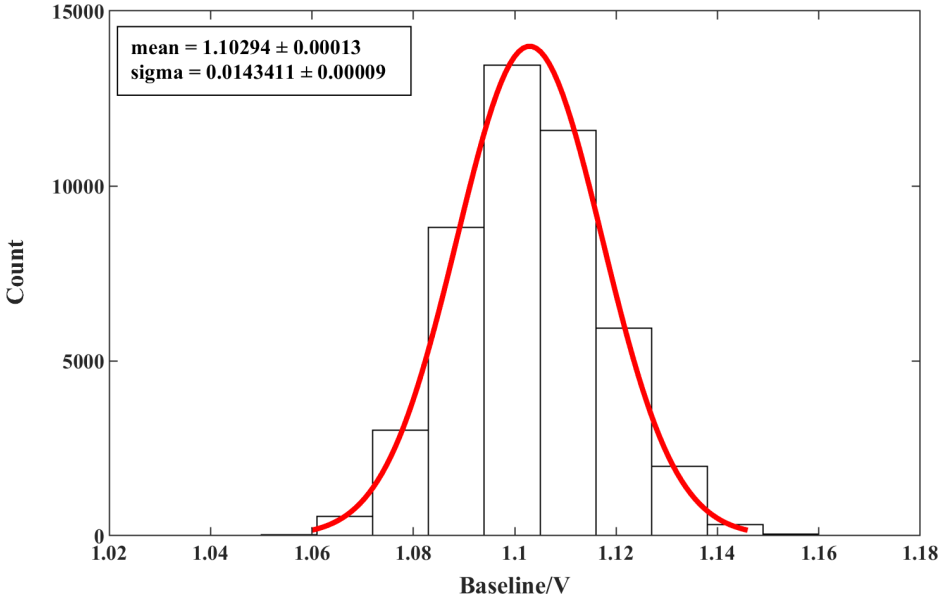


Figure 5. The distribution of the baseline.

3.3 Gain and nonlinearity

Figure 6 exhibits the output amplitude changes with input charge. The expected input range in NνDEx-100 is less than 40 ke^- [11] and demanding a nonlinearity of less than 1%. Based on the measured data, the CSA features an output range from 0 to 6.64 fC (about 41.5 ke^-) with a gain of 220 mV/fC, and the nonlinearity of the entire readout path including the CSA and the unity-gain buffer is less than 1%.

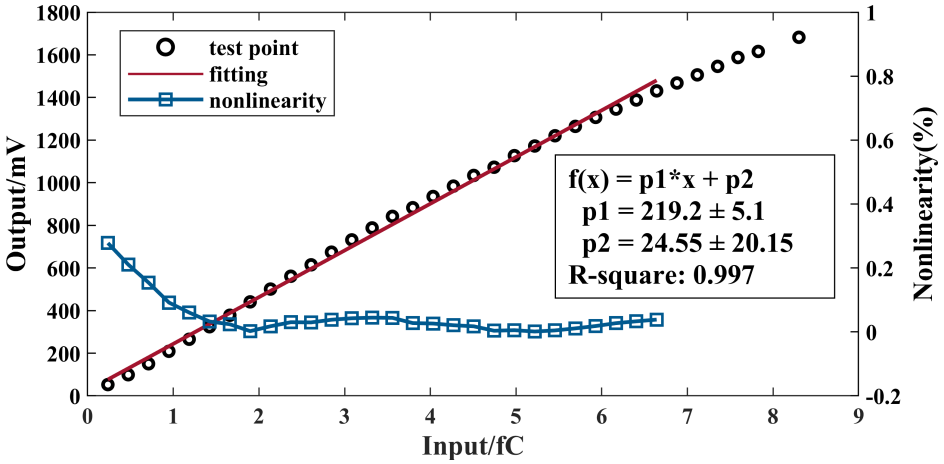


Figure 6. Test results of gain and linearity.

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4 Conclusion

A low-noise Topmetal-S fabricated in a 130 nm CMOS process has been designed and tested for gain-less charge readout in high-pressure TPC. The input linear dynamic range of the CSA is about 6.64 fC and the charge-conversion gain is about 220 mV/fC. The ENC is about 115 e^- after a digital trapezoidal pulse shaper.

Acknowledgments

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